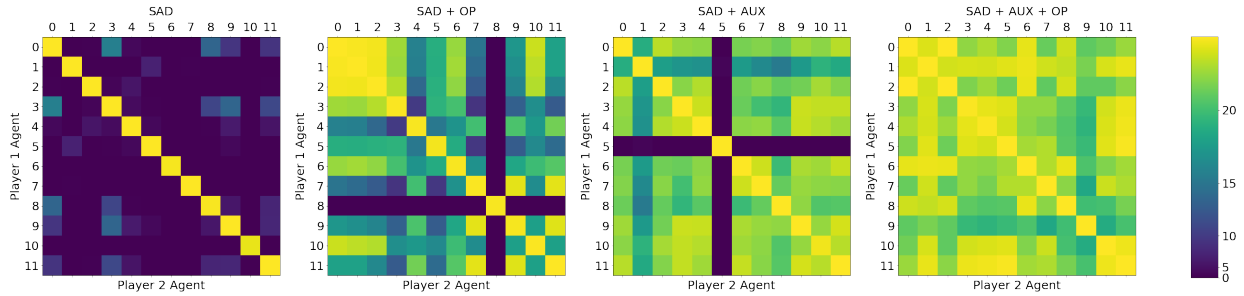

Figure 4. Cross-Play Matrix. Visualization of paired evaluation of different agents trained under the same method. y-axis represents the agent index of player 1 (first mover) and x-axis represents that of player 2. Agents 0-2: 1-layer FC, without residual connection; Agents 3-5: 1-layer FC, with residual connection; Agents 6-8: 2-layer FC, without residual connection; Agents 9-11: 2-layer FC, with residual connection. All agents have 2-layer LSTMs after the FC layers. Each block in the grid is obtained by evaluating the pair on 10K games with different seeds. Please refer to Table 1 for numeric results.

Method	Cross-Play	Cross-Play(*)	Self-Play
SAD	2.52 ± 0.34	3.02 ± 0.39	23.97 ± 0.04
SAD + OP	15.32 ± 0.65	18.28 ± 0.36	23.93 ± 0.02
SAD + AUX	17.65 ± 0.69	21.09 ± 0.18	$\mathbf{24.09 \pm 0.03}$
SAD + AUX + OP	$\mathbf{22.07 \pm 0.11}$	$\mathbf{22.49 \pm 0.18}$	24.06 ± 0.02

Table 1. Cross-Play Performance. The average performance of pairs of agents that are train with the same method but different network architecture and/or seeds. Please refer to Figure 4 for visualization of performance for each individual pair. Cross-Play score is non-diagonal mean of each grid. Cross-Play(*) is the cross-play score after removing the worst model from the grid. Self-Play score is the score attained when agents play with the partner they are trained with.

subsequent rank hint. This is not a typical strategy played by seasoned human players but is easy understand and thus makes the agent easier to cooperate with. However, due to the way we implement OP in Hanabi, it is still possible to form secretive conventions such as using all color hints to indicate a specific move. For example, the worst partner of SAD + OP uses all color hints to indicate playing the 5th card.

6.5. Hanabi with Humans

So far we have focused on AI agents that play other AI agents and have shown that OP is a meta-equilibrium with respect to learning rules in the zero-shot setting.

We now ask: do human strategies in Hanabi also have an OP-like quality? In other words, do OP agents perform well with humans?

To begin to answer this question we recruited 20 individuals from a board game club. These individuals were familiar with Hanabi but not expert players. We asked each individual to play a game of Hanabi with two bots, in random order, using the user interface open-sourced by (Lerer et al., 2019). We note that we did not provide the participants with any information about the bots, either regarding their strategy or the method through which they were trained.

For testing we selected our best SAD + AUX + OP agent based on the cross-play performance (henceforth OP bot). We also have individuals play with the SOTA self-play agent from (Hu & Foerster, 2019) (henceforth SP bot). We download models from their GitHub repo and pick the model based on cross-play scores. For reference, the SP model used here gets 23.99 in self-play and 20.99 in cross-play with other agents where the only difference among them is seed.

Since in Hanabi the exact deck being used can make a huge difference (for example, some hands are unwinnable), to reduce the variance of our results we play each seed by two different players, one for our OP agent, and one for the control. Importantly, to prevent any adaptation advantages, we alternate the order between which bot came first across different participants.

Humans achieved an average score of 15.75 (s.e.m. 1.23) with the OP bot and "bombed out" 45% of games. Thus the OP bot, which has high cross-play scores with other OP bots is also able to play with humans. Note, in our counting convention players keep the current score when they bomb out, which we believe is more appropriate for the zero-shot setting.

By comparison, humans paired with the SP bot achieve an average score of 9.15 (s.e.m. 1.18) and an 85% bomb

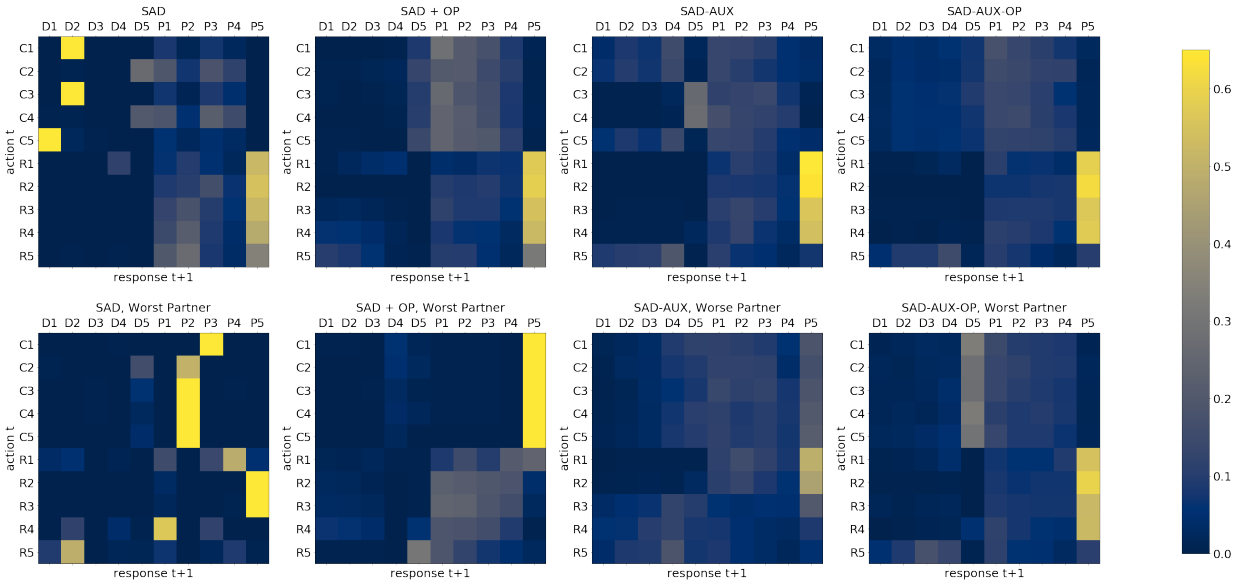

Figure 5. $P(a_t^i \mid a_{t-1}^j)$ Matrices. Each subplot can be used as a proxy to roughly decipher conventions of an agent. The y-axis means the action taken at time-step t and x-axis means the percentage of each action as response at time-step $t+1$. Here we only show a quarter of the matrix that corresponds to the interaction between color/rank hint and play/discard positions. C1-C5 and R1-R5 mean hinting 5 different colors and ranks respectively while D1-D5 and P1-P5 mean discarding and playing 1st-5th cards of the hand. For each plot, we take an agent and run 1000 episodes of self-play to compute statistics. The agents that achieved the highest cross-play scores in Figure 4 are used to generate the top row and their worst partners are chosen to render the bottom row.

rate. To the best of our knowledge, the only other research involving human-AI collaboration in Hanabi is (Eger et al., 2017). Here, a hand-coded AI agent, designed to play well with humans is used and achieves an average of around 15.0 points when paired with humans. Beyond the average scores and "bomb-out" rates, we thus also have access to pairwise comparisons for the two bots when playing two different people on the same deck.

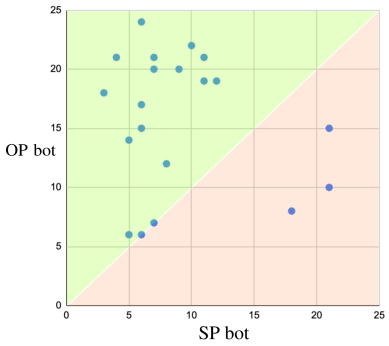

Figure 6. Shown are all scores obtained by human testing. Each blue dot is one seed, when humans are matched with the SP bot (x-axis) and the OP bot (y-axis).

These preliminary numbers confirm that our OP bot significantly outperformed the SOTA self-play bot from (Hu & Foerster, 2019) when paired with humans. In particular, OP won 15 out of the 20 per-seed comparisons and tied in 2 cases, losing to the control group for 3 seeds (p=0.004[1]).

Of course, these results do not suggest that OP will work in *every* zero-shot coordination where AI agents need to cooperate with humans. However, they are encouraging and suggest that OP is a fruitful research direction for the important problem of human-AI coordination.

7. Other Attempts

While attempting to make progress on the zero-shot coordination problem in Hanabi we tried a variety of approaches. Here we discuss some other approaches that seemed promising but did not yield agents that were able to coordinate with agents they were not trained with. While this does not necessarily mean these approaches are doomed to failure, we report these results as information for other researchers interested in this problem.

In particular we tried multi-agent RL adaptations of cognitive hierarchies (Stahl, 1993), k-level reasoning (Costa-Gomes et al., 2001) and training a population of agents. Our original inspiration was that both cognitive hierarchies and k-level reasoning should reduce the tendency towards arbitrary symmetry breaking and have been shown to pro-

[1] exact binomial test of the null hypothesis being that P(OP higher score) $\leq$ P(control higher score).

duce human like decision making in other settings (Wright & Leyton-Brown, 2010). Similarly, population based approaches are gaining popularity for regularizing communication protocols in the field of emergent communication, see e.g. (Fitzgerald, 2019; Tieleman et al., 2018; Lowe et al., 2019). We found that none of these approaches produced high cross-play performance in Hanabi, which we now consider a necessary condition for high zero-shot performance with humans. In hind-sight, considering that all of these approaches would necessarily fail in the matrix game example, this is not at all surprising. Still, to help future researchers learn from our endeavours, we are adding all results to the supplementary material and will open-source the corresponding agents.

8. Conclusion

We have shown that a simple expansion of self-play which we call *other-play* can construct agents that are better able to zero-shot coordinate with partners they have not seen before. We have proven theoretical properties of the OP strategy, shown how to implement it with deep RL, and shown in experiments with the cooperative card game Hanabi that OP can construct robust agents that can play with other AIs as well as with humans.

We do not claim that OP is a silver bullet for all zero-shot coordination problems. In particular, because OP is a modification of the SP algorithm, it can be combined with many of the algorithmic innovations that have been developed to improve SP in various games (Lanctot et al., 2017; Foerster et al., 2018a; Lowe et al., 2017; Foerster et al., 2017). Thus, we believe that this represents an exciting research direction for those interested in moving deep RL beyond two-player, zero-sum environments to ones involving coordination and cooperation. Currently we are assuming that the symmetries Φ are given to the algorithm. However, in principle, discovering the symmetries of an MDP is another optimization problem, which opens interesting avenues for future work.

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